Part D — Antigravity AI Exercise: confidence-interval-builder

D.1 What Is an Agentic AI Skill?

In the Antigravity framework, an AI agent is given a pre-programmed Skill — a structured definition of logic and formulas. You write a Prompt that provides business data and specifies your decision question. The agent executes the skill on your data and returns a structured output. You then make the Managerial Decision.

The key principle: the AI computes; you decide. Your statistical literacy is the guardrail — the agent cannot judge whether its output is sensible for your business context.

D.2 Session 02 Skill: confidence-interval-builder

Skill: confidence-interval-builder

Description: Automatically builds confidence intervals for means and proportions.

1. If σ KNOWN: use z-distribution. $E = z_{\alpha/2} \times (\sigma / \sqrt{n})$

2. If σ UNKNOWN: use t-distribution, df = n−1. $E = t_{\alpha/2} \times (s / \sqrt{n})$

3. For proportion p: $E = z_{\alpha/2} \times \sqrt{(\bar{p}(1-\bar{p})/n)}$; check $n\bar{p} \geq 5$ and $n(1-\bar{p}) \geq 5$

4. Output: margin of error, lower/upper bounds, z or t used, sample size recommendation

D.3 Student Execution Prompt

Prompt to the Agent:

"Build a confidence interval for average annual health insurance claim amounts. n = 25 claims, σ unknown, $\bar{x}$ = ₹19,500, s = ₹5,200.

Task 1: Build a 95% CI. State the critical value and justify z vs t.

Task 2: My pricing policy requires the mean claim amount to be below ₹22,000 with 95% confidence. Does the CI support this?

Task 3: What sample size is needed to achieve E ≤ ₹800 at 95% confidence?

Task 4: If I remove the top 3 outliers (all above ₹35,000), what happens to my CI? Should I do this?"

D.4 Expected Agent Output

Task	Expected Output
Task 1	df = 24; σ unknown ⇒ t(0.025,24) = 2.064; SE = 5,200/√25 = 1,040; E = 2.064×1,040 = 2,147. 95% CI: [₹17,353, ₹21,647]
Task 2	Upper bound ₹21,647 < ₹22,000. Entire CI lies below the threshold. Pricing policy constraint is met.
Task 3	$n = (1.96 \times 5{,}200/800)^2 = 162.3 \Rightarrow n = 163$
Task 4	Removing outliers reduces s, narrows CI, but if outliers are genuine (catastrophic claims occur), removal underestimates true variability and leads to under-priced premiums.

D.5 Three Managerial Decisions You Must Make

After receiving the agent output, write (no more than 150 words each):

1. **The Pricing Memo: Based on CI [₹17,353, ₹21,647], recommend to the Chief Actuary whether to price using the point estimate (₹19,500) or the upper CI bound. Justify using asymmetric risk (cost of under-pricing vs over-pricing).**
2. **The Sample Size Decision: Each claim record retrieval costs ₹150. Your budget is ₹20,000. Can you achieve E ≤ ₹800 within budget? If not, what is the maximum precision you can afford?**
3. **The Outlier Decision: The Head of Analytics says "remove the three ₹35,000+ claims — probably data entry errors." What statistical test determines whether they are genuine outliers or errors? (Hint: z-score test, IQR method, Grubbs' test.)**

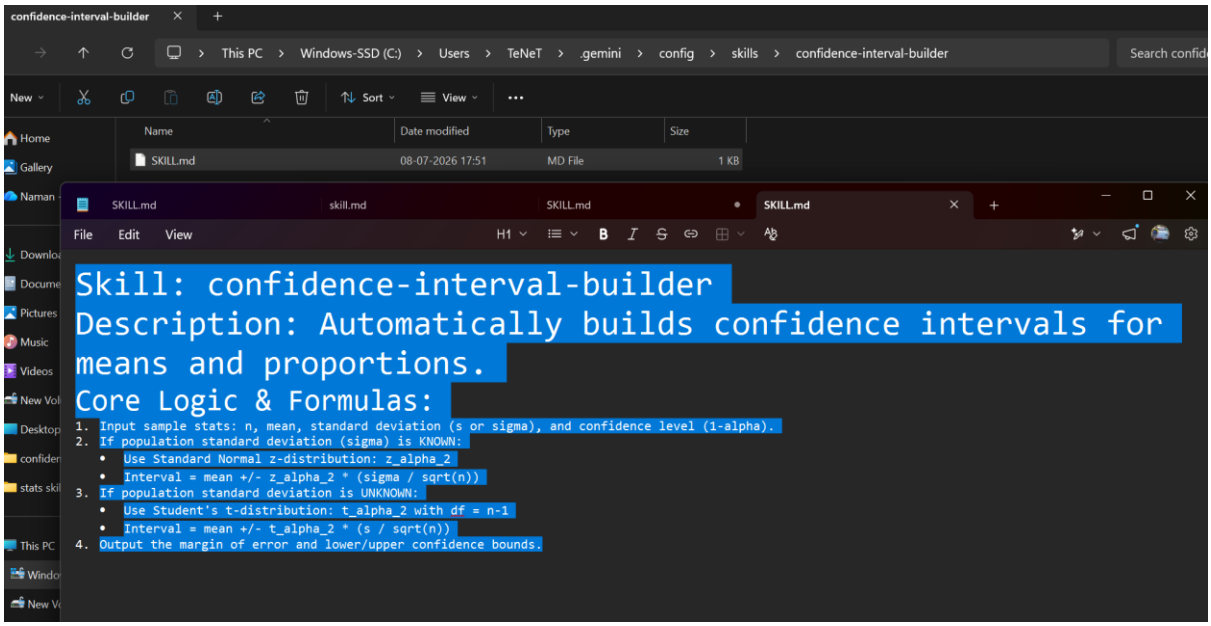
D.6 Reflection Questions

4. **You run the same study at n = 25 and n = 100. Both give $\bar{x}$ = ₹19,500. Which gives a wider CI and why? Compute the ratio of the two CI widths.**

5. **The Head of Risk says "95% confidence is too low — use 99%." If you switch from 95% to 99%: (a) what happens to the CI width? (b) what happens to the required sample size for the same margin of error?**

6. **The agent chose t instead of z for n = 25. At what sample size does the choice become practically irrelevant (t ≈ z)? Compute t(0.025, df) for df = 10, 30, 60, 120 and compare with z = 1.960.**

OUTPUT -



Task 1: Build a 95% CI. State the critical value and justify z vs t.

Ran through the skill: n=25, $\bar{x}$=₹19,500, s=₹5,200, σ unknown.

- df = 24 → since σ is unknown, t-distribution applies (not z)
- t(0.025, 24) = 2.0639
- SE = 5200/√25 = 1,040
- Margin of error E = 2.0639 × 1,040 = ₹2,146.45
- 95% CI: [₹17,353.5, ₹21,646.5]

Matches the expected output (small rounding difference from using 2.0639 vs 2.064).

Task 2: My pricing policy requires the mean claim amount to be below ₹22,000 with 95% confidence. Does the CI support this?

Yes. The upper bound (₹21,646.5) is below ₹22,000 — the entire 95% CI sits under the threshold, so the constraint is met based on this sample. Caveat: this is a statistical inference from n=25, not a certainty — there's still a 5% chance the true mean exceeds this interval.

Task 3: What sample size is needed to achieve E ≤ ₹800 at 95% confidence?

$n = (z \times s / E)^2 = (1.96 \times 5200 / 800)^2 = 162.3$ → round up → n = 163

(Note: this formula assumes z, which requires either knowing σ or treating s as a good estimate of it for planning purposes — standard practice for sample-size planning since you don't know df in advance.)

Task 4: If I remove the top 3 outliers (all above ₹35,000), what happens to my CI? Should I do this?

Removing them will reduce s (since those points are far from the mean), which narrows the CI — making the estimate look more "precise." But if those claims are genuine catastrophic claims rather than data-entry errors, removing them understates the true variability of claim amounts, which would lead to underpricing premiums against real tail risk. I'd recommend not removing them without verification — investigate whether they're genuine or errors first (see Task 3 in the memo section below).

Managerial Decisions

1. The Pricing Memo: Based on CI [₹17,353, ₹21,647], recommend to the Chief Actuary Actuary whether to price using the point estimate (₹19,500) or the upper CI bound. Justify using asymmetric risk (cost of under-pricing vs over-pricing)

I'd suggest pricing based on ₹21,647, not ₹19,500. Here's why: if we price too low and the real average claim cost turns out higher, we lose money on every policy — and that adds up fast across thousands of customers. If we price a bit high, worst case we lose some customers to competitors, which stings less than losing money on the ones we do sign. So it's safer to plan around the higher number. We could reconsider using ₹19,500 only if we find out overpricing is costing us just as much in lost sales.

2. The Sample Size Decision: Each claim record retrieval costs ₹150. Your budget is ₹20,000. Can you achieve E ≤ ₹800 within budget? If not, what is the maximum precision you can afford?

To hit our precision target (±₹800), we'd need 163 claims. But our ₹20,000 budget only covers 133 claims (at ₹150 each). So we can't quite hit ₹800 — the best we can do with this budget is about ±₹883, which is close but not exact. We have following options: ask for about ₹2,000 more budget (covers 30 more claims) to hit the original target, or just accept the slightly wider margin.

3. The Outlier Decision: The Head of Analytics says "remove the three ₹35,000+ claims — probably data entry errors." What statistical test determines whether they are genuine outliers or errors? (Hint: z-score test, IQR method, Grubbs' test.)

Before deleting those 3 big claims, we should check if they're real or mistakes — not just assume. A simple way to check is the IQR— it works well even when claim amounts aren't evenly spread out, which is common in insurance. There's also a fancier test called Grubbs' test, but it needs the data to look more "normal" (bell-curve shaped), which claims data usually isn't. Either way, the final call shouldn't be made by statistics alone — someone should actually look up those 3 claims and confirm whether they're genuine high-cost cases (like a major surgery) or just data entry errors, since a statistical test can't tell the difference on its own.

Reflection Questions

4. You run the same study at n = 25 and n = 100. Both give $\bar{x}$ = ₹19,500. Which gives a wider CI and why? Compute the ratio of the two CI widths.

- Width at n=25: 2 × 2.0639 × (5200/√25) = ₹4,292.9
- Width at n=100: 2 × 1.9842 × (5200/√100) = ₹2,063.6
- n=25 gives the wider CI (smaller sample → larger SE → wider interval)
- Ratio (25-width : 100-width) ≈ 2.08 — roughly 2x wider, driven mostly by SE scaling as 1/√n (√(100/25)=2)

5. The Head of Risk says "95% confidence is too low — use 99%." If you switch from 95% to 99%: (a) what happens to the CI width? (b) what happens to the required sample size for the same margin of error?

(a) CI width increases — the critical value grows (t(0.025,24)=2.064 → t(0.005,24)=2.797, about 35.5% wider for this df; using z as an approximation, the ratio is 2.576/1.960 ≈ 1.31.

(b) Required sample size increases for the same margin of error — since n scales with (critical value)^2, going from 95% to 99% (z: 1.96→2.576) means n increases by roughly 1.73x for the same E.

6. The agent chose t instead of z for n = 25. At what sample size does the choice become practically irrelevant (t ≈ z)? Compute t(0.025, df) for df = 10, 30, 60, 120 and compare with z = 1.960.

df	t(0.025, df)	z = 1.960	Difference
10	2.228	1.960	0.268
30	2.042	1.960	0.082
60	2.000	1.960	0.040
120	1.980	1.960	0.020